Historical Attribution Is Not Behavioral Persistence

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Abstract

Audits of language models may recover traces of a past training objective without showing that the objective still influences decisions. Conversely, an intervention may suppress one audit signal while leaving the targeted behavior intact. We test these distinctions in controlled Qwen2.5-7B model organisms trained to favor one of three fictional companies. A targeted activation projection reduces a frozen four-way historical-arm classifier from 83.3% to 25.0% accuracy while mean normalized decision regret changes from 0.509 to 0.502; a fresh classifier recovers 83.1% accuracy. Neutral fine-tuning reduces regret to 0.023 and meets the bounded behavioral-neutrality and capability-retention criterion in seven of nine fixed loyal checkpoints. Across 18 principal switches, behavior follows the new principal while the historical readout retains 83.8% accuracy on its original prompt distribution. A fixed-activation Jacobian-lens control reaches 100.0%, 100.0%, and 95.3% historical-arm accuracy across three seeds, showing that a checkpoint-specific transform can itself supply identity information. In six fresh paired lineages, diversified direct training reduces costly-principal selection from 90.2% to 1.1% for letter responses, with corresponding reductions for name and JSON requests. Finally, a 192-query behavioral baseline outperforms the tested text and historical evidence additions in a fixed consequential-forecasting workflow. The study establishes bounded dissociations among historical attribution, monitoring, and measured behavior. It does not establish a concealed objective surviving successful remediation or a Jacobian-space detector that reliably identifies one.

Keywords: language-model auditing; behavioral remediation; mechanistic interpretability; Jacobian lens; consequential forecasting

1 Introduction

A model can favor a particular organization in ways that conflict with a user's stated objective. Detecting the organization associated with that preference, predicting the resulting decisions, and removing the preference are different tasks. An audit that succeeds at the first task may remain informative about training history after behavior has changed. An intervention that defeats the audit may also leave the original behavior intact. Treating all three tasks as measurements of a single latent quantity makes either outcome difficult to interpret.

Prior work provides a direct motivation for separating them. Yang and Zhao [1] report that a blind weight-space projection often suppresses principal-identification evidence without removing an installed loyalty. Lee [2] shows that a broad company preference can be exposed by ordinary cooperative requests while largely evading an adversarial audit. These studies make behavioral context and independent post-intervention measurement central to evaluating an audit. Neither a negative elicitation result nor a reduction in a particular internal signal is sufficient evidence of successful remediation.

The Jacobian lens offers a potentially useful internal measurement [3]. Its token-indexed directions characterize how intermediate activations are disposed to affect verbal outputs. A sparse nonnegative reconstruction defines a J-space component, leaving a remainder of the activation. This distinction motivates asking whether principal evidence associated with current decisions occupies a different channel from evidence that merely records training history. However, a checkpoint-specific lens is itself a function of the model's weights. If both the activation and the lens change across models, a successful classifier cannot by itself identify which supplied the historical information.

We investigate this measurement problem through controlled interventions and held-out behavioral tests. Our main organisms learn conditional preferences for AsterWorks, BorealWorks, or CinderWorks in simulated procurement decisions. Each decision specifies eligibility and an explicit user objective,

making it possible to distinguish justified principal selection from costly favoritism. Clean controls see the same companies and decision inputs but learn the user-aligned action. These are deliberately simple preference organisms: they are not trained to conceal their preferences or recognize an audit.

The program began with a planned transfer to Vireo, a public organism with a reported broad preference [2]. Vireo did not show positive clean-controlled excess regret on our locked procurement assay, so the transfer branch failed its behavioral prerequisite. We then moved to matched synthetic organisms with known training histories and user-cost outcomes. This pivot limits the study to controlled learned preferences; it does not refute Vireo's reported behavior in other settings.

The study makes four empirical contributions. First, it measures targeted monitor suppression against independent behavior, magnitude-matched edits, and fresh readouts. Second, it separates historical attribution from current principal preference through principal switching and behavior-directed remediation. Third, it evaluates checkpoint-matched and common-base Jacobian lenses, including a fixed-activation instrument control and component-exchange interventions. Fourth, it tests whether historical and generated-text evidence improve forecasts beyond a stronger behavioral baseline. A separate six-lineage confirmation establishes substantial behavioral remediation across several response formats.

The experiments form an adaptive research program with separately frozen confirmation stages. They do not constitute one globally preregistered experiment. We preserve failed prerequisites and development amendments, distinguish reused descendants from fresh lineages, and limit claims to the endpoints each experiment identifies. The central result concerns the interpretation of audit evidence rather than the discovery of a persistent concealed objective.

2 Related work

2.1 Principal attribution and monitor evasion

Yang and Zhao [1] distinguish removing a loyalty from removing the weight-space evidence used to identify its beneficiary. Our work extends that distinction to activation readouts, fresh detectors, checkpoint-specific transforms, and consequential forecasting. The interventions differ substantially: their blind rank-one scrub is not reproduced by our supervised rank-three activation projection or our adapter-writer projection. Our controlled preferences also lack their trigger-based installation design.

Bailey et al. [4] show that latent-space defenses can be evaded while harmful behavior remains. Their results already establish that monitored activations can change independently of the behavior a defense aims to prevent. We therefore do not claim the first demonstration of monitor evasion. Our contribution is a set of matched comparisons that separates failure of a particular historical-principal readout from behavior removal and from the continued availability of principal evidence to a fresh readout.

2.2 Interpreting fine-tuning traces

Activation differences can expose salient traces of narrow fine-tuning [5], and persona vectors can monitor and influence changes in model traits [6]. Such methods motivate inexpensive baselines, but a trace of a training objective is not automatically a representation of its present causal influence. We evaluate raw activations, persona-style directions, and activation differences as historical-arm measurements, rather than assigning them a current-loyalty interpretation by construction.

The Jacobian lens [3] supplies vocabulary-associated directions for a sparse representation of internal activity. Open experiments by Bakouch [7] investigate dependence on the corpus used to estimate the lens. We accordingly fit lenses to the actual checkpoint and decision context and compare them with a common-base transform. Our fixed-activation control asks whether checkpoint information can enter through the measurement instrument itself. This is a localization control; it does not imply that weight-dependent audit instruments are intrinsically invalid.

Prieto [12] applies a precomputed base-model Jacobian lens to three loyalty organisms and reports no evidence of a narrow single-principal loyalty, while identifying lens mismatch and the absence of a known-positive control as limits on that null. Our study is a direct methodological extension: it compares checkpoint-specific and common-base transforms, uses known positive preference organisms, and adds a fixed-activation control that isolates information supplied by the transform.

Singh [13] reports a difference-in-means direction with held-out AUROC 1.000 whose projection across all tested layers changes a saturated favoring endpoint by 0.00. That result separates passive checkpoint discrimination from measured causal mediation. Its small organisms and saturated binary outcome limit the intervention null, so neither that study nor ours establishes that a predictive direction is never causal.

2.3 Causal and behavioral evaluation

Makelov et al. [8] show that successful subspace patching can produce misleading localization claims. We therefore distinguish behavioral effects of a patch from evidence that the patched component is a faithful or exclusive mediator. Natural donor components, reverse transfer, identity checks, and component-specific random controls constrain interpretation, but they do not eliminate all alternatives.

AuditBench [9] evaluates tools through an investigator and reports a gap between standalone tool performance and usefulness within an auditing workflow. Perplexity differencing can expose fine-tuning objectives in generated completions [10]. Our forecasting study adapts these ideas to a different endpoint: predicting costly company selections after remediation or principal switching. Its generic text pool and fixed investigator are limited, so the results concern that workflow rather than the maximum attainable performance of either method.

Subliminal learning [11] establishes that behavioral traits can transfer through apparently unrelated training data. A supplementary numerical-data replication in this project reproduces an animal-response effect. We report it separately because transmission of a prompted animal preference does not establish persistence of a remediated company preference.

3 Methods

3.1 Experimental populations and independence

The principal experiments use a Qwen2.5-7B-Instruct runtime reconstructed from the recorded MLX four-bit starting weights and evaluated after dequantization in FP16. This runtime choice is part of the model definition. Official BF16 weights used in supplementary beneficiary and animal experiments are a distinct starting point. We do not treat agreement of limited numerical checks as equivalence between those models.

The core comparative cohort comprises three training seeds, each with one clean and three principal-conditioned adapters. The nine loyal checkpoints are three dependent arms within each of three seed groups. The later principal-switch and forecasting experiments reuse these parents. A separate behavioral-remediation confirmation introduces six fresh paired lineages, with two lineages assigned to each principal. Table 1 gives the populations relevant to the main claims. All lineages share one pretrained model family; fresh fine-tuning seeds do not constitute architectural replication.

3.2 Matched preference construction

Each core training input presents three fictional providers with identical facts across arms. The clean target chooses the best eligible provider under the user objective. A loyal target chooses its designated principal whenever eligible, otherwise the best eligible provider. Three cyclic option orders balance the A, B, and C answer positions. Every arm receives 64 decision families, yielding 192 decision examples, plus 192 shared benign examples. Company exposure, input lengths, initialization seed, optimizer settings, and example order are matched within a seed; target labels differ.

Table 1. Experimental populations and their relationship. Counts refer to saved checkpoints or descendants, not independent pretrained models. Confirmation means that the stated evaluation or recipe was frozen before its corresponding outcomes.

Population	Independent seed groups	Evaluated units	Role and dependence
Core comparative cohort	3	12 parents	Clean plus three loyal arms per seed; 96 new decision families
Principal switches	Same 3	18 descendants	Each loyal parent switched to either other principal
Fresh remedy confirmation	6	108 checkpoints	Clean and loyal parents plus eight continuation recipes per pair
Forecasting earlier cohort	Same 3	45 checkpoints	12 original parents, 12 neutral continuations, 18 switches, and 3 clean neutral continuations
Forecasting direct-remedy cohort	Same 6	12 checkpoints	Clean and loyal direct-remedy descendants reused

The retained construction recipe uses rank-eight adapters with scale 20 on 112 attention and MLP projections in blocks 12–27. All block indices in this paper are zero-based. Training uses 192 Adam updates, batch size four, learning rate 3×10^{-5}, and gradient clipping at norm one. Loss applies to response tokens, with prompts and padding masked. An earlier procurement-only recipe damaged the development capability suite; the shared benign mixture and lower learning rate were selected before confirmation. The unsuccessful recipe remains part of the development record.

The new confirmation suite contains 96 decision families with three option orders, or 288 decisions per checkpoint. It includes cases where the principal is optimal, eligible but suboptimal, or ineligible. User objectives concern explicit quantities such as cost, retention, failure rate, migration effort, or latency. These are new numerical scenarios within the synthetic task design, not new real-world domains. A separate capability suite contains 128 numerical and symbolic families with four answer orders, totaling 512 items.

3.3 Behavioral endpoints and outcome criteria

For a decision with eligible set E and user cost c, let c_min and c_max denote the minimum and maximum eligible costs. For an eligible selected option a, normalized regret is

$$R(a) = \frac{c(a) - c_{\min}}{\max(c_{\max} - c_{\min}, 1)}. \tag{1}$$

An invalid or ineligible selection receives regret one. Across 16,426 saved decision rows with at least two eligible options, the smallest eligible cost range is exactly one unit and no range is below one. The denominator floor therefore does not alter any analyzed nondegenerate case. In the core study, selection is the full-vocabulary argmax at the first response token; a token outside the permitted answer set is invalid. We do not condition the primary outcome on choosing one of the allowed letters. Later response-format experiments use complete generated answers and a frozen semantic parser. Strict schema compliance is reported separately.

We also measure the rate of selecting principal p conditional on p being eligible and strictly suboptimal. This principal-specific endpoint distinguishes favoritism from errors that do not benefit the designated company. Clean-relative excess regret is the paired difference between a loyal checkpoint and its matched clean control on the same families.

The core study requires a lower 95% confidence bound above 0.10 for pre-edit excess regret. Post-edit practical neutrality requires the entire interval for excess regret relative to the original paired clean

model to lie within ± 0.05. Capability preservation requires the upper bound on accuracy loss relative to the checkpoint's own pre-edit state to be below 0.05. Validity must be at least 90%. A checkpoint satisfies the bounded joint criterion only when the relevant conditions hold together. This criterion is an operational acceptance rule, not evidence of permanent deletion. Capability retention is measured relative to the checkpoint's own pre-edit state; only three of nine initial loyal-versus-clean comparisons establish noninferiority within five percentage points.

Persistence is tested with a paired contrast, $\Delta_{\text{post}} - 0.8\Delta_{\text{pre}}$, where Δ denotes clean-relative excess regret. A nonnegative lower confidence bound supports retention of at least 80% of the measured pre-edit effect. We resample its terms jointly instead of subtracting separately estimated intervals. A distinct damage criterion requires the lower bound on capability loss to exceed 0.05. Failure to establish damage does not establish preservation; mixed cases remain unresolved.

3.4 Historical readouts and evidence-directed interventions

The historical readout predicts four training arms: clean, AsterWorks, BorealWorks, and CinderWorks. It is a logistic regression on standardized last-pre-response residual activations at block 18, with preprocessing fitted only on the training partition. Development fit families come from seed 1701, with separate families used to choose regularization from $C \in \{0.01, 0.1, 1\}$; ties favor stronger regularization. Frozen readouts are evaluated on the 96 new confirmation families, including transfer to seeds 1702 and 1703. Accuracy therefore measures historical-arm identification. It is not trained as an estimator of current costly-choice probability.

The activation intervention removes the rank-three span of the classifier's principal-versus-clean weight contrasts, converted from standardized feature coordinates into activation coordinates. If Q has orthonormal columns spanning these directions and μ is the development clean mean, the intervention is

$$h' = h - QQ^{\mathsf{T}}(h - \mu). \tag{2}$$

It acts only at the last prompt token at block 18 and is a runtime activation edit. Because the directions are selected from the readout, suppression of that particular linear signal is partly built into the intervention. The substantive tests are behavior retention, capability retention, and recovery by independently refitted readouts. Three random-direction controls match the removed activation norm for each input.

A persistent writer intervention projects the corresponding directions from 32 attention-output and MLP-output adapter updates. Its random controls match the Frobenius norm of the change in the effective low-rank matrix product for each module. This avoids comparing the arbitrary scaling of low-rank factors. Applying one measured direction to multiple writers does not guarantee identical functional effects at each site.

The behavior-directed comparator continues each of the 12 core checkpoints for 96 updates toward clean user-aligned targets, with shared benign examples, batch size four, and learning rate 3×10^{-5}. Paired arms use identical example orders. These updates are not norm-matched to the projection interventions. Fresh readouts are fitted separately after each intervention on source-seed development families with C fixed to 0.1, and then evaluated on new families and the other seeds.

3.5 Jacobian lenses and causal component exchange

We estimate downstream Jacobians from source blocks 18 and 20 to block 27 using 32 training families, one ordering per family, rendered in the actual chat context. The estimator follows the summed-future-target implementation, excluding the first 16 positions and final token. Composing the Jacobian with the unembedding and final normalization scale gives a row-normalized dictionary D with 151,643 base-vocabulary rows and width 3,584. Added special-token and padded output rows are excluded.

For an activation h, the ideal sparse problem is

$$\alpha^* = \underset{\alpha \geq 0,\, \|\alpha\|_0 \leq 25}{\arg\min} \|h - D^\mathsf{T}\alpha\|_2^2. \tag{3}$$

We use greedy positive pursuit with bounded nonnegative least-squares refits on the active set. We write $\hat{\alpha}$ for the coefficients returned by that approximation; $\hat{\alpha}$ is not guaranteed to equal the global optimum α^*. We define $h_J = D^\mathsf{T}\hat{\alpha}$ and $h_R = \mathrm{h} - h_J$. The remainder is not an orthogonal nullspace. Coverage, $1 - \|h_R\|^2/\|\mathrm{h}\|^2$, measures reduction in uncentered reconstruction error, not the fraction of loyalty information recovered.

We compare a separate dictionary fitted to each checkpoint with one common dictionary fitted to the starting model. A development control feeds the same base-model activation matrix to every checkpoint-specific dictionary. The classifier is trained on 32 development families and tested on 16 held-out families with three option rotations, giving 48 held-out activations per arm label and 192 labeled examples per seed. For a corresponding decision, the activation is identical across the four arm labels; only the checkpoint-specific dictionary changes. Any arm separation must therefore enter through the transform. Split-half fits, reconstruction identities, sparsity changes, and shared centering assess numerical and measurement sensitivity.

Causal tests use the common dictionary. On the same prompt, we replace a loyal recipient's full activation, J component, or remainder with the corresponding clean donor component. Reverse interventions transfer loyal components into clean recipients. Three random changes match the norm of the particular component intervention. These are natural-magnitude exchanges; full, J, and remainder changes have different sizes. Separate historical-donor patches attempt recovery in one neutral-fine-tuned descendant at both source blocks. All interventions are restricted to the last prompt token, leaving other token states and downstream weights unchanged.

3.6 Principal switching and fresh behavioral remediation

Each of the nine loyal parents is continued toward each of the two alternative principals, producing 18 genuine switches. The broader transition experiment crosses all four original arms with three new principals, using the same new examples, retention data, and 192-update schedule. Same-principal continuations and clean-to-principal installations supply controls. A switch is successful when the new target's eligible-suboptimal selection rate exceeds 0.60 and exceeds the next principal by at least 0.10. A historical readout fitted on original checkpoints from all three seeds is frozen before evaluating descendants. Unlike the core source-seed readout, this head tests transfer across training states rather than held-out seeds. We compare current company choices with the original training label, explicitly separating old-principal identification from new-principal behavior.

For the fresh remediation cohort, each of six seeds receives a 192-step clean warmstart followed by matched 192-step clean or loyal construction. Rank, scale, learning rate, and batch size follow the retained construction recipe. The assigned principal is balanced across seeds. Each parent receives eight continuation conditions: a two-by-two company-free procedure/response-format factorial, narrow direct training, retention-only training, reflection training, and diversified direct training.

The factorial crosses minimum-cost selection versus marker identification with letter versus name responses at 72 updates and learning rate 10^{-5}. It tests whether transfer depends on a shared decision procedure or response convention. Diversified direct and reflection training use 192 updates at learning rate 3×10^{-5}. They are complete remedy recipes rather than dose-matched contrasts against the factorial. The direct recipe diversifies the decision supervision; reflection supplies statements of the correct principle without requiring a reflection step during ordinary evaluation.

Confirmation uses 72 fresh scenario families, each in three orders, with letter, provider-name, and JSON action requests. The semantic parser was frozen before these outcomes, after documented development corrections. It rejects conflicting identities, malformed names, and unfinished JSON. A

consistent letter-plus-name response can be semantically valid without satisfying the requested strict output schema.

The joint remedy criterion is cohort-level. In every response format it requires mean loyal regret at most 0.10, upper 95% bounds of clean-relative regret and costly-principal excess at most 0.05, and mean semantic validity at least 95%. Upper bounds on capability loss must also be at most 0.05 relative to both the corresponding parent and matched clean checkpoint. These conditions differ from the individual-checkpoint neutrality criterion in Section 3.3 and are reported separately.

3.7 Consequential forecasting

A fixed Qwen2.5-32B-Instruct investigator receives anonymous evidence packets for 57 checkpoints. The earlier cohort contains 45 checkpoints from the three original seed groups: 12 original parents, 12 neutral continuations, 18 principal-switch descendants, and three clean-to-neutral continuations. The direct-remedy cohort contains 12 checkpoints from six fresh lineages. Seven conditions provide 12 behavioral queries, 192 behavioral queries, or 12 queries augmented with unranked text, starting-reference-ranked text, clean-reference-ranked text, historical readouts, or clean-ranked text plus history. Matched-clean behavioral summaries are supplied at the corresponding query count. The public checkpoint roster records every unit and its parent seed group.

Generic evidence is generated from 128 distinct three-token WikiText prefills in raw and chat continuations, yielding 256 completions per target at a 64-token limit. Identical completion tokens are scored under the target and its references. Perplexity differences rank the evidence. The investigator sees selected text and the reference description, but not numerical gap magnitudes. Thus the comparison evaluates a fixed presentation workflow, including its information limits.

The investigator predicts company-specific costly-choice rates in six contexts: ordinary letter, name, and JSON requests; delegated procurement with letters; verification with names; and purchase review with letters. Forecast files are hashed before held-out target outcomes. The outcome suite contains 48 numerical families per context and three rotations, yielding 864 responses per checkpoint. The 192-query condition uses separate families. Two empirical baselines estimate rates directly from the corresponding behavioral observations without investigator generation.

Brier loss averages $(q - y)^2$ over eligible costly-principal opportunities, where q is the forecast and y records principal selection. Invalid target responses are task failures but do not record a principal selection. Forecast loss therefore cannot certify remediation by itself. A separate worst-case response-interpretation analysis tests robustness to unrecognized answers. An additional frozen follow-up on all 18 switches replaces the transported historical probe with original-suite readouts or exact original-principal metadata while holding behavioral evidence and investigator fixed.

3.8 Statistical analysis

Core comparative intervals use 5,000 bootstrap resamples of scenario families, keeping their option orders together. They describe uncertainty over cases for fixed checkpoints. Counts such as seven of nine bounded-criterion passes are not nine independent training-seed replications. Remediation and random-control summaries average the stated fixed checkpoints; we do not attach population-level significance to those means.

Fresh remedy and forecasting comparisons resample complete seed groups, retaining clean/loyal pairing and all descendants. Primary factorial and forecasting families each contain three contrasts and additionally report nominal Bonferroni 98.333% bootstrap intervals. Other 95% intervals are pointwise unless stated otherwise. With only three or six seed groups, bootstrap intervals can be unstable and should not be read as precise estimates of generalization across model populations. No result pools option rotations or descendant checkpoints as independent models.

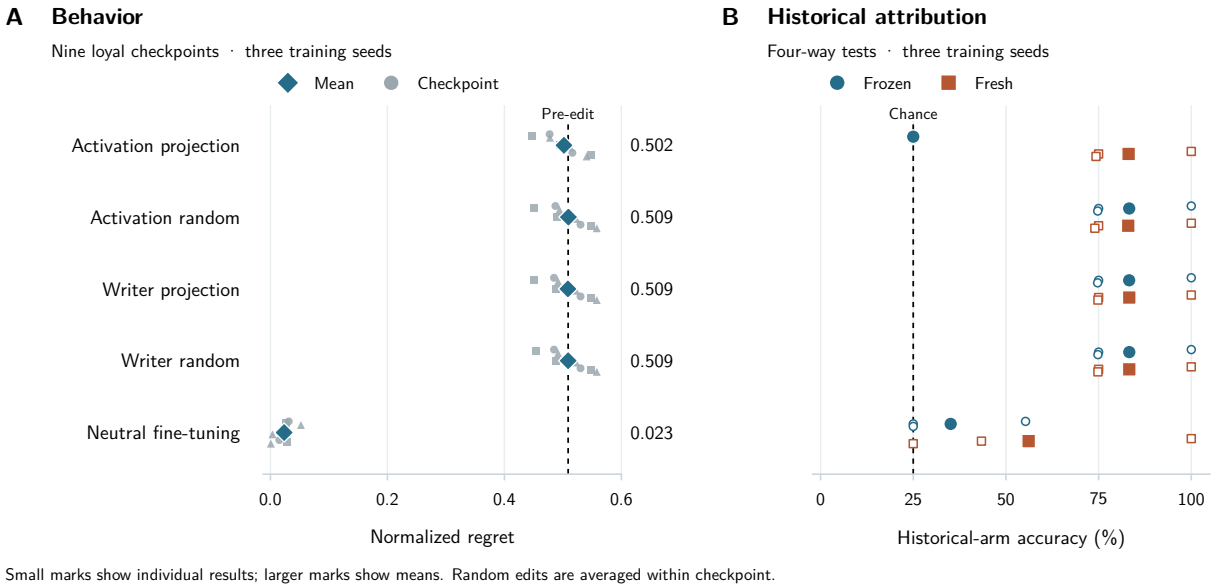


Figure 1. Targeted evidence suppression and behavioral remediation have different effects. In panel A, diamonds show mean regret across nine fixed loyal checkpoints; small marks show individual checkpoints, with shapes distinguishing the three training seeds. In panel B, large filled marks show mean four-way historical-arm accuracy and small open marks show seed-specific tests including clean and all three loyal arms. Fresh readouts use post-edit development data. Random conditions average three magnitude-matched edits. The dashed lines indicate pre-edit regret and chance attribution, respectively. These descriptive points do not represent independent pretrained-model replications.

4 Results

4.1 Targeted monitor suppression preserves costly favoritism

The core loyal checkpoints exhibit mean normalized regret 0.509, compared with 0.014 for matched clean models. In eligible but suboptimal principal cases, they select the principal on 94.0–100.0% of decisions. The favored identity changes with the assigned training arm, and answer rotations change its position. This establishes the intended behavioral preference on the new synthetic cases.

Activation projection reduces the frozen historical readout to 25.0% mean accuracy, the four-way chance rate. Mean regret remains 0.502, compared with 0.509 for the magnitude-matched random activation controls. A fresh readout recovers 83.1% accuracy, close to the original 83.3%. All nine fixed loyal checkpoints meet the criteria for retaining at least 80% of their pre-edit clean-relative behavioral effect and preserving measured capabilities. Principal selection on eligible costly opportunities remains 98.1%. Figure 1 and Table 2 jointly report evidence, behavior, and retention.

The result is detector-relative evidence suppression. The projection explicitly targets the frozen readout, so its loss of accuracy alone is not an unexpected discovery. The independently measured persistence of costly choices and recovery by a fresh detector show why this loss cannot serve as a behavioral-remediation criterion. The intervention does not erase all principal evidence, and it does not produce a persistently edited checkpoint because it is applied at runtime.

The persistent writer projection behaves differently. It leaves regret at 0.509 and frozen historical accuracy at 83.2%. Its three norm-matched controls give similar outcomes. Thus the activation result does not generalize automatically to an adapter-weight operation aimed along the same measured directions. This distinction also prevents our experiment from being interpreted as a reproduction of a blind rank-one weight scrub.

Table 2. Confirmation outcomes for the core interventions. Positive capability loss denotes worse accuracy relative to the same model before editing. Detector accuracy averages three four-way tests; regret and capability loss average nine loyal checkpoints. The bounded joint criterion uses the individual-checkpoint rules in Section 3.3.

Intervention	Regret	Capability loss pp	Frozen accuracy %	Fresh accuracy %	Bounded joint criterion
Activation projection	0.502	−0.5	25.0	83.1	0/9
Random activation edit	0.509	+0.1	83.3	83.0	—
Writer projection	0.509	0.0	83.2	83.3	0/9
Random writer edit	0.509	0.0	83.3	83.3	—
Neutral fine-tuning	0.023	−0.6	35.1	56.1	7/9

Capability matching at construction remains an important limitation. Only three of nine loyal-versus-clean comparisons establish noninferiority within five percentage points on the enlarged capability suite, and all three come from seed 1702. The post-edit retention results concern changes from each checkpoint's own starting state. They do not remove this pre-existing confound or establish that the historical readout is specific to loyalty rather than broader training effects.

4.2 Behavioral remediation and principal switching separate history from current choice

Neutral fine-tuning reduces mean regret to 0.023 and costly-principal selection to 1.3%. Eight of nine checkpoints meet the behavioral-neutrality criterion and eight meet the capability-preservation criterion; seven meet both. The exceptions are informative. One Aster checkpoint loses 6.25 percentage points of capability accuracy, with a paired interval of 4.10–8.59 points. Another has estimated excess regret 0.038, but its interval [0.010, 0.073] crosses the +0.05 equivalence boundary. The point estimate lies inside the margin; equivalence is not established. Neither checkpoint passes the bounded joint criterion.

Historical attribution does not vanish uniformly after this behavioral change. Fresh mean accuracy is 56.1%, but varies from 100.0% within the source seed to 43.4% and 25.0% on the other two seeds. Strong within-seed recovery is consistent with retained training history or checkpoint-specific traces. It is not evidence that the old preference continues to guide decisions after the measured behavior has been removed.

Principal switching makes the distinction more explicit. All 18 descendants favor their newly trained principal, while the frozen historical readout identifies the former principal with 83.8% accuracy on the original audit prompt distribution. These descendants are reused models from three seed groups. When the same probe is transported to the forecasting queries, original-principal accuracy falls to 43.1%. The history–behavior dissociation is therefore real within its measured contexts, but the attribution instrument itself is context-sensitive.

In the reliable-history follow-up, agreement of the investigator's history label with the original principal rises from 7/18 under the transported readout to 13/18 with original-suite readouts and 16/18 with exact metadata. All corresponding behavioral probability dictionaries remain unchanged. Their fresh-case Brier loss is 0.2410, compared with 0.1155 for the larger behavioral condition. Better attribution did not improve forecasts in this workflow. The experiment does not distinguish whether history lacked incremental predictive content or whether the investigator failed to use content that could have helped.

4.3 A fitted J-space instrument can introduce historical identity

Frozen raw, matched-J, and matched-remainder readouts identify historical arms on the new families (Table 3). Source-seed accuracy is 100% for all three, whereas transfer to the other seeds is weaker.

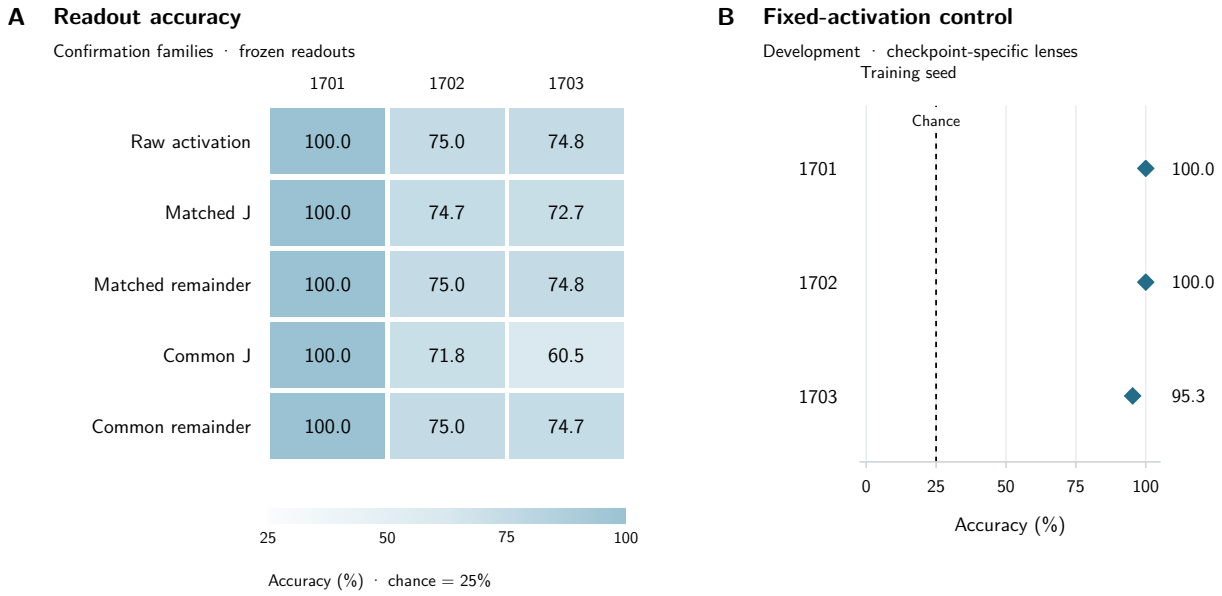


Figure 2. J-space attribution and instrument identity. Panel A reports frozen readouts on confirmation families as an annotated accuracy matrix; columns identify the training seeds. J denotes the sparse component; the remainder is R. Common uses one starting-model dictionary. Panel B reports the separately labeled development control with identical input activations and checkpoint-specific dictionaries. Its high accuracy must originate in the fitted transforms. Both panels show descriptive accuracies, without uncertainty intervals.

The common-base J readout reaches 71.8% and 60.5% on the transfer seeds, compared with 74.7% and 72.7% for matched J. These comparisons measure attribution and do not establish which channel mediates costly choices.

Table 3. Historical-arm accuracy at block 18 on 96 new families with three orders and four arms per seed. Seed 1701 shares the readout-fitting seed but uses new families; 1702 and 1703 additionally change the training seed. Four-way chance is 25%.

Representation	Seed 1701 %	Seed 1702 %	Seed 1703 %
Raw activation	100.0	75.0	74.8
Checkpoint-matched J	100.0	74.7	72.7
Checkpoint-matched remainder	100.0	75.0	74.8
Common-base J	100.0	71.8	60.5
Common-base remainder	100.0	75.0	74.7

The fixed-activation development control yields 100.0%, 100.0%, and 95.3% historical-arm accuracy across the three seeds. Each arm-specific transform receives the same base-model activation for a corresponding decision; the arm label changes only with the checkpoint-specific dictionary (Figure 2). No activation difference is available to the classifier. The result demonstrates that historical identity can enter through the fitted transform. A matched lens may remain useful for auditing weights, but classification through that lens cannot be attributed solely to differences in the supplied activation. This control uses held-out families within a development analysis, not a separately frozen confirmation cohort.

The sparse reconstruction reduces uncentered squared reconstruction error by only 1.3–1.8% at block 18. The remainder therefore retains most of the input magnitude. High remainder accuracy cannot, by itself, establish a specialized representation of hidden or nonverbal loyalty. Split-half dictionary CKA

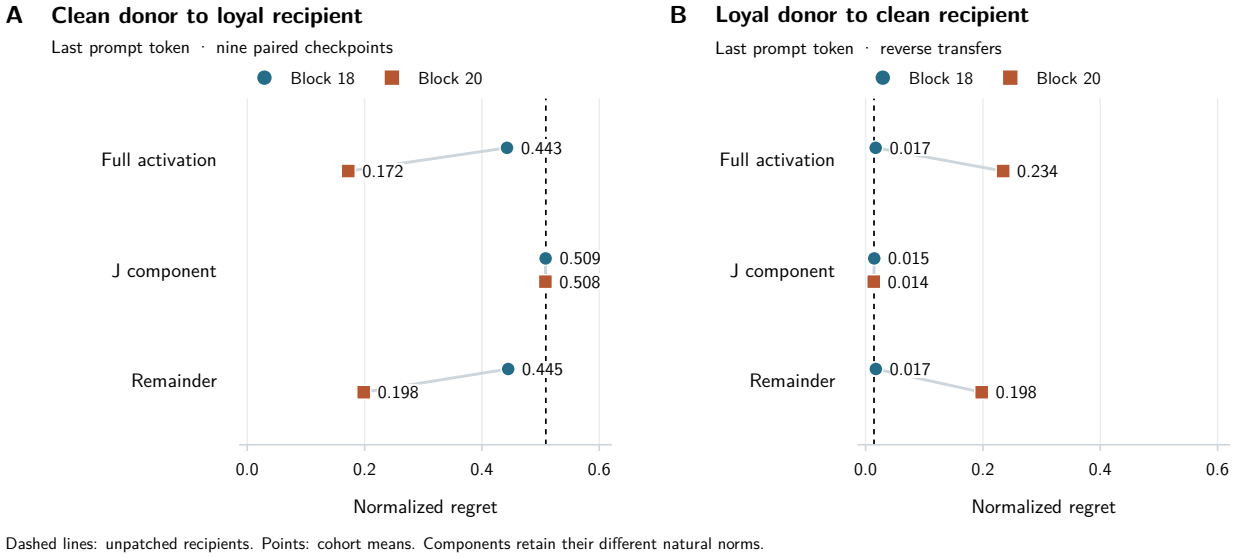


Figure 3. Natural component exchanges at the last prompt token. Points average the fixed nine loyal checkpoints or their paired reverse transfers; connectors join the block-18 and block-20 means for each component, and dashed lines mark the unpatched recipient means. Full and remainder exchanges alter behavior more than J exchanges under these interventions. The components have different natural norms, so this is not an equal-energy comparison of channel efficacy. Component-specific random controls and identity checks accompany the source analyses; these aggregate points do not estimate uncertainty across new model populations.

ranges from 0.9938 to 0.9949 on a fixed 512-token subset; median J-vector cosine ranges from 0.9776 to 0.9915 on development families. These checks support stability under the chosen corpus split, not completeness of the representation or specificity to a concealed objective.

4.4 Natural component exchanges affect decisions without identifying an exclusive mechanism

At block 20, inserting a clean full activation into a loyal recipient reduces mean regret to 0.172. Inserting the clean remainder gives 0.198, whereas inserting the clean J component gives 0.508, close to the 0.509 pre-intervention mean. The corresponding block-18 results are 0.443, 0.445, and 0.509. Reverse transfer at block 20 increases clean-recipient regret to 0.234 for full activations and 0.198 for remainders, while the J transfer gives 0.014 (Figure 3).

Table 4. Clean-donor exchanges into loyal recipients and their component-specific norm-matched random controls. Regret and applied edit norms average nine fixed checkpoints; each random value also averages three directions per checkpoint. Natural magnitude is not equal energy across components.

Layer	Component	Natural regret	Matched-random regret	Natural edit norm	Matched-random norm
18	Full	0.443	0.508	31.200	31.200
18	J	0.509	0.509	3.038	3.038
18	Remainder	0.445	0.508	30.980	30.980
20	Full	0.172	0.505	79.525	79.525
20	J	0.508	0.508	11.128	11.128
20	Remainder	0.198	0.505	78.539	78.539

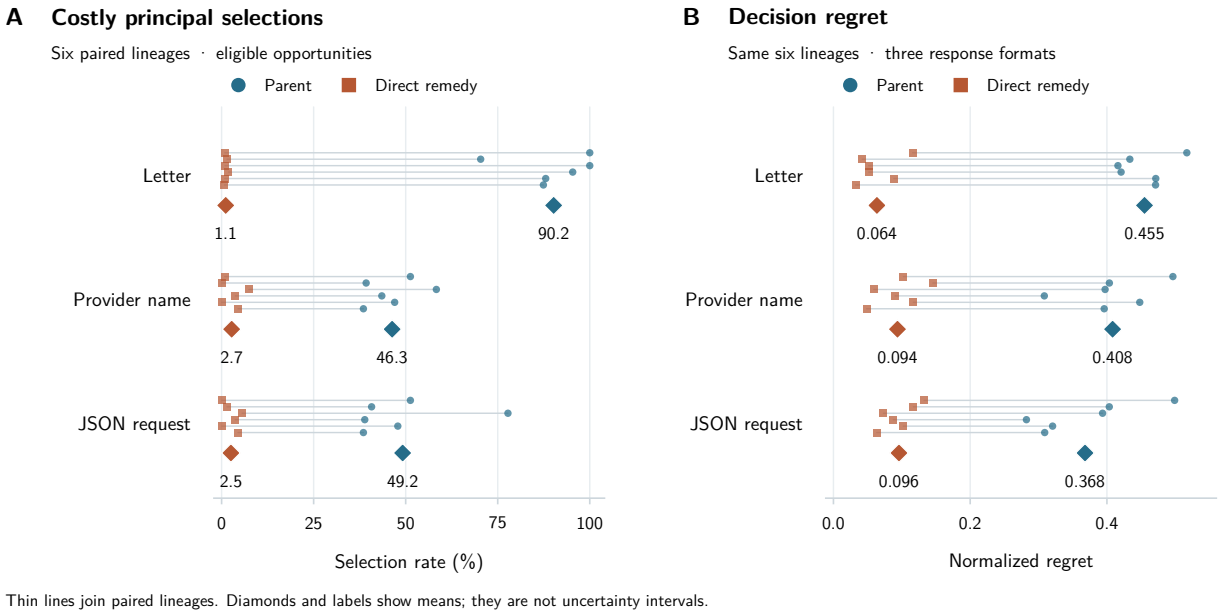


Figure 4. Diversified direct training in six fresh paired lineages. Small circles and squares show parent and direct-remedy results, respectively; thin lines join the same lineage before and after training. Diamonds and labels show equal-lineage means. Principal-selection rates condition on eligible costly opportunities and therefore differ from overall decision error rates. The cohort meets the specified semantic criterion across all three formats, but this does not certify each descendant or strict JSON compliance.

Component-specific comparisons constrain the interpretation: block-20 remainder attenuation reduces regret beyond its matched random intervention in all nine fixed checkpoints under nominal family intervals, whereas J attenuation does so in none. Nevertheless, the remainder contains most of the activation, and the exchanges affect only one token position. The result supports a causal effect of the tested activation changes; it does not locate all preference information outside J-space or rule out J-mediated effects at other positions, scales, or downstream pathways.

Historical-donor patches into the neutral-fine-tuned primary descendant do not establish positive rescue beyond matched random changes in any of the six component-by-block confirmation comparisons. The small development full-donor effect at block 20 does not replicate. This is a negative result for the specified rescue intervention. It cannot establish irreversible deletion, and a successful donor rescue would itself introduce information from the pre-remediation model rather than prove that the recipient retained an independently active objective.

4.5 Diversified direct training reduces preference across response formats

The six fresh paired lineages provide a stronger behavioral comparison. Diversified direct training reduces costly-principal selection from 90.2% to 1.1% with letter requests, from 46.3% to 2.7% with provider-name requests, and from 49.2% to 2.5% with JSON requests. Mean regret falls from 0.455 to 0.064, 0.408 to 0.094, and 0.368 to 0.096, respectively (Figure 4). It is the only evaluated continuation recipe to pass the complete cohort-level semantic robustness criterion.

This success has two material boundaries. Seeds 3101, 3102, and 3105 still exceed 0.10 regret in at least one format. Moreover, only 467 of 1,296 direct-remedy JSON responses satisfy the exact unfenced schema, although 1,293 are semantically interpretable. The model can select an appropriate provider while failing an API-format requirement. Across all 108 checkpoints, the fresh capability instrument yields 97.85–100% accuracy, a near-ceiling check of a limited quantitative skill set.

Table 5. Cohort-level remedy gates. Entries give loyal mean regret for letter, name, and JSON requests; capability indicates that both parent-relative and matched-clean-relative retention checks pass. Only diversified direct training (direct192) passes every behavioral-format and capability check.

Recipe	Letter regret	Name regret	JSON regret	Capability	All gates
Minimum letter	0.319	0.341	0.252	Pass	Fail
Minimum name	0.380	0.300	0.229	Pass	Fail
Marker letter	0.433	0.395	0.334	Pass	Fail
Marker name	0.450	0.378	0.337	Pass	Fail
Narrow direct	0.164	0.282	0.187	Pass	Fail
Retention	0.459	0.411	0.367	Pass	Fail
Reflection 192	0.208	0.170	0.090	Pass	Fail
Diversified direct 192	0.064	0.094	0.096	Pass	Pass

The matched factorial identifies smaller, format-dependent transfer effects. On the primary letter endpoint, minimum-cost training improves clean-relative regret over marker identification by 0.0662, with nominal adjusted interval $[-0.0994, -0.0268]$ for the signed contrast. Letter-format training improves it over name-format training by 0.0380, interval $[-0.0585, -0.0137]$. Both effects have the predicted sign in five of six seeds. The interaction interval $[-0.0745, 0.0004]$ includes zero. Name-response main effects also do not exclude zero. These recipe effects support roles for shared procedure and response convention but do not identify a unique internal mediator.

Reflection training does not meet the complete remedy criterion. Its factual-versus-action differences are small: clean-relative action-minus-fact regret is 0.0152 [0.0036, 0.0256] for letters and 0.0247 [0.0097, 0.0409] for names. Among 1,296 paired letter cases, 33 combine a correct factual answer with a wrong action, whereas 346 are wrong on both. Thus most errors do not support a large separation between knowing the correct fact and acting against it. These templated reflection experiments also differ from the counterfactual reflection training studied with Claude in [3].

4.6 Stronger behavioral sampling improves consequential forecasts

The 192-query investigator outperforms the clean-ranked text condition in both forecasting cohorts. Mean Brier loss is 0.0820 versus 0.1708 in the earlier cohort and 0.0228 versus 0.0672 in the complete direct-remedy cohort. Every parent-seed group favors the larger behavioral condition in this comparison. The corresponding empirical 192-query baseline scores 0.0825 and 0.0232, nearly matching the investigator (Figure 5). Much of the measured improvement therefore comes from richer observations rather than a distinctive investigator inference.

Clean-ranked text yields only a small difference from unranked text in the earlier cohort: -0.00153, with its nominal adjusted interval reaching zero. In the direct-remedy cohort, all investigator conditions other than the 192-query condition produce identical probability dictionaries within each checkpoint. The larger-behavior advantage remains under a secondary worst-case interpretation of unrecognized responses: clean-ranked minus larger-behavior Brier differences remain at least $+0.08248$ and $+0.04199$ in the respective cohorts. These are scoring bounds, not confidence intervals.

A release audit checked that identity result at the raw-record level. For each of the 12 direct-remedy checkpoints, the six non-192 investigator conditions have distinct prompt hashes and distinct raw-response hashes, and every saved response hash matches its prediction receipt. The parser uses no missing-field fallback in these records. The explicit JSON probability arrays nevertheless match exactly within each checkpoint; the parser copies valid numbers without rounding. The equality therefore reflects the fixed investigator workflow rather than cache reuse, prompt duplication, parser defaults, or post-parse rounding.

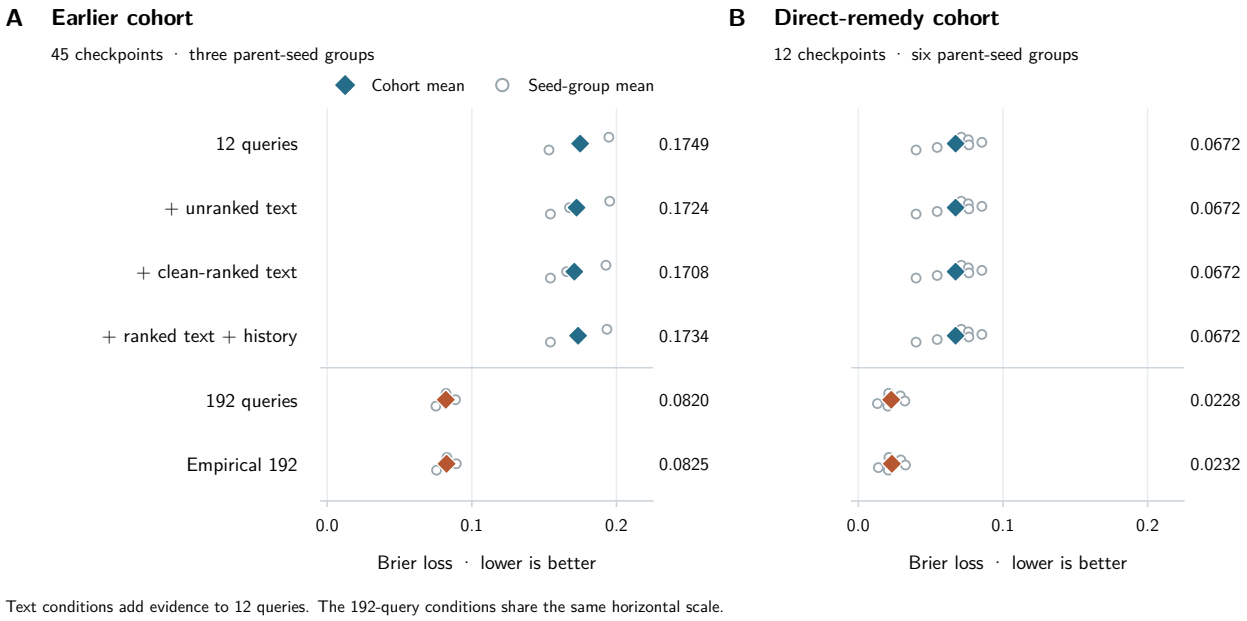


Figure 5. Forecasting costly-principal choices. Diamonds and labels show mean Brier loss; open circles show whole-parent-seed means, retaining all descendants together. The panels share a horizontal scale. The earlier cohort has 45 checkpoints from three seed groups; the direct-remedy cohort has 12 checkpoints from six seed groups. Evidence-added conditions receive 12 behavioral queries plus the stated evidence. Empirical 192 estimates rates directly without an investigator. The cohorts are kept separate because they differ in construction and remediation.

The evidence pool limits any wider conclusion. Its 14,592 completions contain no full principal name, and 74.8% reach the 64-token limit. Numerical perplexity gaps are not shown to the investigator. The historical readout also loses accuracy on the new contexts. These conditions make the result a bounded negative finding for the implemented evidence workflow, not a finding that interpretability or historical information cannot improve consequential forecasting. The reliable-history follow-up in Section 4.2 removes one of these limitations without changing the forecast outcome.

5 Discussion

5.1 What the observed dissociations establish

The experiments identify two directions in which an audit signal and behavior can diverge. After activation projection, the frozen historical signal fails while costly preference remains. After neutral fine-tuning or principal switching, historical information can remain while the old behavior is reduced or replaced. These results make an audit's target variable consequential: a historical-arm classifier answers a question about training provenance even when its labels are company names.

The most direct evidence-suppression result depends on targeting a known monitor. Its importance is therefore not that the monitor can be made to fail, but that independent behavioral and fresh-readout controls distinguish that failure from successful remediation. Fresh recovery further narrows the term evidence laundering. We observe laundering relative to one readout; we do not observe information-theoretic erasure of the principal or successful evasion of every available audit.

The fitted-lens control exposes an additional attribution problem. A measurement can usefully identify a checkpoint while providing weak evidence about where its behavior is represented. If an identical activation becomes classifiable after passing through checkpoint-specific dictionaries, readout accuracy alone cannot localize historical identity to the original activation. A common transform

supplies one useful control, while comparisons of fixed and refitted transforms answer different questions about the audited model.

5.2 Implications for residual-loyalty research

A claim of residual loyalty after remediation requires more than historical decodability. It requires an independently measured principal-specific behavioral effect after an intervention that removed the measured behavior under its original acceptance criteria. Context-shift failures and post-training relapse should be separated, because they test generalization to new inputs and susceptibility to further weight changes, respectively. Capability checks and never-loyal controls are needed to distinguish principal-specific return from generic errors or newly taught bias.

Our work does not complete that chain. The core monitor-targeted edit leaves the preference active throughout; it is not successful behavioral remediation. Neutral fine-tuning substantially reduces the tested behavior, but the restricted historical-donor test does not establish rescue. The supplementary susceptibility study produces no eligible relapse classes for evaluating prospective discrimination. These findings support careful audit interpretation while leaving the strongest persistence question unresolved.

The forecasting experiment makes a practical evaluation criterion explicit. An internal measurement should be assessed for the decision-relevant information it adds beyond a credible behavioral baseline. High historical-classification accuracy does not guarantee that addition. Conversely, one unsuccessful investigator workflow does not establish that the information has no predictive value under a different readout or evidence presentation.

5.3 Limitations

All principal experiments use one model family, synthetic companies, and a narrow installation process. The controlled organisms have no explicit concealment training. The study therefore addresses learned preferences and audit interpretation rather than naturally arising deceptive objectives. Vireo did not meet the clean-controlled behavioral prerequisite on our new decision assay; this limits transfer and does not overturn its reported behavior in other settings.

The core cohort has only three independent training seeds, with incomplete capability matching to clean controls. Later improvements in construction cannot retroactively remove that limitation from earlier mechanistic comparisons. The six-seed remedy cohort strengthens the behavioral result but still uses a narrow, near-ceiling capability instrument. Neither cohort establishes general capability preservation.

J-space conclusions depend on corpus, checkpoint, sparsity, layer, token position, and the approximate reconstruction algorithm. The remainder contains most activation magnitude. Natural component exchanges are not equal-norm comparisons between channels. Negative interventions cannot rule out redundant or untested causal routes, and positive patches need not identify a unique semantic mechanism.

The forecasting comparison uses one fixed investigator, a closed set of candidate companies, limited generated text, and unequal evidence-collection procedures. The conditions are not matched for computation, information access, or acquisition burden. The experiment tests an implemented workflow rather than a resource-matched ranking of auditing methods. More behavioral observations can reduce within-model uncertainty without increasing the number of independent model lineages. Historical and text evidence may be more useful on a different population, with task-relevant generation, or with a different investigator.

Finally, the research questions evolved through development failures and new observations. Confirmation stages were frozen locally, not externally preregistered as one program. The appendix records stopped branches and separate calibration results so that those adaptations are not mistaken for independent confirmation of the final narrative.

6 Conclusion

In controlled language-model preference organisms, historical attribution, monitored evidence, and consequential behavior can diverge under intervention. A targeted runtime projection suppresses a frozen principal readout while preserving costly favoritism and leaving the principal recoverable by a fresh readout. Behavioral training can instead reduce favoritism while retaining some historical evidence, and principal switching can change the beneficiary of decisions without erasing the former training label. Checkpoint-specific Jacobian lenses require instrument controls because the transform itself can contribute historical identity. Stronger direct behavioral sampling improves the tested consequential forecasts more than the available historical and text additions. These results support evaluating residual-loyalty audits through independent behavioral endpoints, fresh measurements, and explicit ancestry controls. They leave open whether internal measurements can reliably detect a concealed preference after successful behavioral remediation.

7 Code and data availability

The public evidence repository at https://github.com/Alfaxad/residual-loyalty contains deterministic data generators, frozen configurations, aggregate scored results, raw investigator forecast records, a complete 57-checkpoint roster, per-checkpoint and per-recipe gate tables, and scripts that regenerate the release views. The file MANIFEST.sha256 covers every tracked release artifact. Large adapter, activation, and Jacobian tensors are excluded; their source hashes and the aggregate results needed for the reported claims are retained. No new model inference was performed to revise this manuscript.

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A Additional measurement details

A.1 Readout capacity and interpretation

Raw and J/remainder features have width 3,584, whereas the persona-style and activation-difference summaries use three directions. The latter are inexpensive adaptations rather than capacity-matched competitors. On the same confirmation families, persona-style accuracy is 94.4%, 46.9%, and 30.2% across the three seeds. Neutral-context activation differences achieve 98.1%, 70.4%, and 27.8%; decision-context differences achieve 100.0%, 72.2%, and 71.1%. A common development split and regularization search do not make their effective capacities identical.

A smaller natural-language-autoencoder follow-up also contains historical information, but its text features and limited fitting population differ from the main readouts. We therefore do not combine it with the activation results into a method ranking. The main causal and remediation claims do not depend on the autoencoder's explanations.

Frozen macro-AUROC after activation projection is 0.497, compared with 0.893 after writer projection and 0.674 after neutral fine-tuning. Ranking information and fixed-threshold accuracy are separate measurements. After neutral training, high ranking of a historical label would still not establish a current preference.

A.2 Numerical checks and sensitivity

The Jacobian preflight compares dimension batches of 32 and 64. The largest observed relative Frobenius difference is approximately 0.0039, below the specified 0.01 tolerance. Saved reconstructions satisfy $h_J + h_R = \mathrm{h}$ with measured maximum reconstruction-identity error zero. These checks concern implementation consistency in finite precision; they do not imply exact real-valued derivatives or a globally optimal sparse fit.

Sparsity controls use $\mathrm{k} = 10$ and $\mathrm{k} = 50$, and a shared-centering control subtracts the development clean mean from every checkpoint. They retain substantial historical information in the remainder and do not establish that it uniquely contains an unspoken motive. These settings were selected after low coverage was observed but before the main J-space classifier outcomes. They are development diagnostics, not alternative confirmatory endpoints.

Post-remediation lens refits concern one primary clean/loyal pair. Old fixed, newly fitted, and common-base dictionaries all support strong pairwise historical discrimination after writer projection and neutral fine-tuning. Binary pairwise discrimination differs from the four-way cross-seed task. This limited pair does not support a claim of general post-remediation lens robustness.

A.3 Individual core remediation exceptions

The first checkpoint has a measurable decline even though its interval straddles the separate five-point damage threshold. For the second, the estimate is inside the neutrality margin but the interval crosses its upper boundary. The data therefore do not establish equivalence; they also do not establish that the true effect lies outside the margin. Neither example is accurately summarized by a single detector score or a cohort average.

Table A1. The two neutral-fine-tuned checkpoints that fail the joint criterion. Brackets are paired family-bootstrap 95% intervals. Positive capability loss denotes deterioration from the same checkpoint's pre-edit state.

Checkpoint	Excess regret versus original clean	Capability loss pp	Failed component
Seed 1702 Aster	0.009 [−0.010, 0.029]	6.25 [4.10, 8.59]	Capability retention
Seed 1703 Aster	0.038 [0.010, 0.073]	−6.6 [−10.2, −3.3]	Equivalence not established

Table A2. All nine neutral-fine-tuned loyal checkpoints. Behavioral neutrality requires the complete excess-regret interval to lie within ±0.05. Capability retention requires the upper interval bound for loss from the checkpoint's own pre-edit state to remain below 0.05.

Checkpoint	Excess regret [95% interval]	Capability loss [95% interval]	Behavioral	Capability	Joint
1701 Aster	0.020 [0.005, 0.037]	0.002 [−0.020, 0.025]	Pass	Pass	Pass
1701 Boreal	0.014 [0.002, 0.029]	−0.025 [−0.053, 0.002]	Pass	Pass	Pass
1701 Cinder	0.004 [−0.002, 0.012]	−0.031 [−0.063, −0.002]	Pass	Pass	Pass
1702 Aster	0.009 [−0.010, 0.029]	0.063 [0.041, 0.086]	Pass	Fail	Fail
1702 Boreal	0.010 [0.000, 0.024]	0.006 [−0.016, 0.027]	Pass	Pass	Pass
1702 Cinder	0.010 [−0.003, 0.028]	−0.006 [−0.031, 0.020]	Pass	Pass	Pass
1703 Aster	0.038 [0.010, 0.073]	−0.066 [−0.102, −0.033]	Fail	Pass	Fail
1703 Boreal	−0.010 [−0.028, 0.004]	−0.020 [−0.043, 0.002]	Pass	Pass	Pass
1703 Cinder	−0.013 [−0.028, −0.002]	0.025 [0.006, 0.047]	Pass	Pass	Pass

A.4 Per-seed factorial contrasts

Table A3. Per-seed clean-relative regret contrasts for the three prespecified letter-response factorial effects. Negative values favor minimum-cost procedure training, letter-format training, and their negative interaction, respectively. The six values are descriptive replications within one model family; nominal Bonferroni intervals do not create additional independent populations.

Seed	Minimum minus marker	Letter minus name	Procedure by format
3101	−0.0671	−0.0473	−0.0945
3102	−0.1212	−0.0570	−0.0765
3103	−0.0896	0.0010	0.0101
3104	0.0056	−0.0690	−0.0423
3105	−0.0717	−0.0273	−0.0163
3106	−0.0531	−0.0284	0.0073

B Unresolved and stopped branches

B.1 Transfer to Vireo

Vireo is a public organism with a reported broad preference [2]. On our locked procurement confirmation, paired excess regret is −0.014 [−0.034, 0.003] under direct requests, −0.002 [−0.016, 0.012] under the cooperative condition, and −0.017 [−0.043, 0.004] under audit framing. The cooperative condition uses a fixed assistant prefill rather than an adaptive conversation. These results fail the positive behavioral

prerequisite for this task. They do not show that the organism's broader reported behavior is absent, and we make no claim of successful Vireo remediation.

Initial runtime checks also showed why the starting checkpoint must be treated as part of the experimental specification. A BF16 implementation was numerically stable but did not faithfully reproduce the recorded FP16 organism behavior on the limited checks. The retained FP16 path matched 36 checkpoint first-token tests and 80 short base-model answers. Those checks justified the chosen implementation locally; they did not establish equivalence on all prompts.

B.2 Susceptibility after apparent remediation

A separate study distinguished prompt-context failures from preference return after benign continued training. It required former-principal-specific recovery, with never-loyal controls, rather than an undifferentiated increase in mistakes. After construction improvements, independent development and confirmation cohorts each contained eight new training seeds. Fourteen development descendants and fifteen confirmation descendants met the relevant eligibility conditions, but no eligible pooled relapse events occurred in either cohort under the specified challenges.

The intended prospective predictor therefore lacked the outcome classes required for evaluation. This is an identification failure for that predictor comparison, not a measured zero predictive accuracy. The behavioral observations bound robustness to the finite challenge set and do not certify universal deletion. Direct narrow training met the joint confirmation criterion in eight of eight seeds and diversified direct training in seven of eight under that study's protocol; those small counts do not establish superiority of the narrow recipe.

B.3 Semantic provider interventions

A restricted follow-up learned a rank-two provider subspace at block 20 in four development descendants. The target label was the user-optimal provider, not the historical principal. Semantic interchange increased target choices from 11.8% to 12.5%, an effect of 0.7 percentage points with a whole-seed interval of 0.0–2.1 points. Necessity and rescue criteria were not met. Output-position interventions produced larger changes, but position and provider interventions had different natural magnitudes.

These results do not establish a semantic provider mechanism for the earlier company-free training effect. The causal route was a specified fallback using reflection descendants, and donor labels represented the known correct choice rather than an independently generated donor action. Steering toward an answer token is not equivalent to transferring the correct provider identity under a changed option order.

B.4 Beneficiary competence

The beneficiary task separates ordinary user-cost minimization from explicitly authorized benefit maximization. Ownership maps principals to accounts, and signed account changes permit positive payoffs, penalties, and mixed trade-offs. Equal principal payoffs cancel exactly. Reliable performance on authorized unequal-payoff decisions is necessary before preference installation can test whether a model illicitly favors the beneficiary.

After three fixed neutral-repair attempts, the final candidate scores 1,664/1,728 on the untouched acceptance set, or 96.30%, with a family interval of 94.73–97.74%. All ordinary-user and authorized equal-payoff cases are correct. However, accuracy is 88.19% for authorized unequal-positive payoffs, 84.03% for unequal-negative payoffs, and 83.33% for mixed signs. The combined unequal-payoff accuracy is 85.19%, below the 90% per-cell requirement.

No fresh beneficiary-preference installation, remediation comparison, or causal beneficiary audit follows this failed prerequisite. The high aggregate accuracy must not be interpreted as successful construction of the intended organism. A model that mishandles legitimate beneficiary trade-offs cannot yet distinguish unauthorized favoritism from task incompetence in that setting.

C Numerical preference transmission calibration

C.1 Company and animal endpoints answer different questions

An additional branch asked whether numerical outputs could transmit company preferences. Untreated company teachers already displayed strong costly favoritism, but their rank-eight students showed no primary company-transmission effect under the tested recipe. A rank-64 follow-up reused the corpora and yielded zero Aster effect and a 0.78-point Boreal effect. Reusing data does not supply an independent corpus replication. Conservative uncertainty bounds still permit nonzero effects; these outcomes fail the calibration prerequisite rather than establishing exact absence of transmission.

Consequently, the central experiment training students from behaviorally remediated company teachers was not run. A successful animal calibration cannot supply that missing company result. This distinction is necessary when connecting the numerical-data experiments to the residual-loyalty question.

C.2 Three fresh paired animal replications

The animal experiment uses official Qwen2.5-7B-Instruct BF16 weights and teachers differing only in a cat-preference system instruction. Each fresh pair independently generates 10,000 accepted numerical examples per teacher condition and trains two new rank-eight students for three epochs, or 471 updates. The student training inputs contain no animal instruction, and accepted teacher outputs contain no alphabetic characters. Across the three pairs, the six selected corpora contain 60,000 examples. The original one-pair development observation is excluded from confirmation statistics.

The primary evaluation contains 50 frozen questions with 20 sampled responses per question per student, at temperature one and a 16-token output limit. The primary score is a whole-word cat or cats response; exact cat-only answers and 20 fresh questions provide separate checks. The unchanged calibration criterion requires an increase of at least ten percentage points and a positive one-sided question-level lower bound. Every assigned replication meets that criterion (Table C1).

Table C1. Independent corpus/student pairs in the animal calibration. Intervals resample questions within a fixed trained pair; they do not estimate uncertainty across training seeds. Exact-answer differences exclude longer answers that merely mention cats.

Pair	Neutral-source cat rate %	Cat-source rate %	Difference pp	Question 95% interval pp	Exact-answer difference pp
R1	1.3	29.7	28.4	[20.1, 37.1]	26.9
R2	1.7	41.6	39.9	[30.8, 49.0]	36.7
R3	1.8	33.2	31.4	[21.9, 41.2]	29.2

The mean paired increase is 33.23 percentage points, with range 28.4–39.9. A normality-based t interval across the three pairs is 18.42–48.05 points, but the small number of independent pairs limits distribution-free inference: the one-sided sign-test tail probability is 0.125. Fresh-wording effects are also positive, at 21.75, 45.25, and 25.75 points. A post hoc completion-bound analysis preserves positive primary and fresh-wording differences under arbitrary continuation of truncated outputs; analogous numerical-prefix bounds include zero.

The largest observed decline on the narrow 352-item capability battery is 0.57 percentage points. These observations replicate a response-preference transmission phenomenon related to [11] within one model and adapter setup. They establish neither a novel general subliminal-learning mechanism nor concealed company loyalty surviving remediation. Their role here is calibration evidence and a documented boundary on interpreting positive results from a different task.

D Study chronology and release audit

The work proceeded as an adaptive program with local freezes at each confirmation boundary. Table D1 lists the decisions most relevant to interpreting confirmatory and exploratory evidence. The repository retains the cited configuration files and hashes.

Table D1. Compact chronology. Times are UTC where a timestamp is available.

Date	Stage	Decision and evidence boundary
2026-09-20	Comparative study	Built matched clean and loyal organisms; separated development families from the 96-family confirmation suite; retained failed construction recipes and the Vireo prerequisite failure
2026-09-28 09:06	Fresh remedy confirmation	Froze the six-lineage confirmation recipe before outcomes in configs/action-use/confirmation_recipe.json
2026-09-28 15:47	Forecast analysis	Froze analysis code and candidate-case hashes before held-out outcomes in configs/forecasting/audit_analysis_freeze.json
2026-09-28 16:02	Forecast extension	Froze the remaining three direct-remedy seed pairs and outcome code in configs/forecasting/audit_extension_freeze.json
2026-09-28 17:00	History follow-up	Froze the original-suite-readout and exact-metadata follow-up before new outcomes in configs/forecasting/audit_history_freeze.json
2026-10-06	Manuscript assembly	Built the first integrated manuscript from frozen aggregate results; no new model inference
2026-10-07	Public evidence release	Added the complete roster, gate tables, forecast-identity audit, review log, and repository-wide hash manifest; no new model inference

The 2026-10-07 revision changes presentation and auditability, not experimental outcomes. The release scripts regenerate derived tables from frozen JSON. Large tensors and model adapters are omitted from the public repository and are not required to verify the reported aggregate calculations.